



Python for Fixed Income

COURSE 1 · PYTHON FOUNDATIONS

Course Project

WEEK 4 · DAY 4

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This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

The scenario: a new month-end extract

Month end, as of 2026-10-30

You are the analyst on the credit desk. Operations has sent the holdings extract, with a cover note.

The portfolio manager wants to know

1. What is in the book
2. Where it is concentrated
3. Whether the file can be trusted
4. How the book differs from its benchmark

A new file. Run the routine, do not remember it.

Everything in the files is invented: the issuers, the tickers, the CUSIPs, the prices, and the positions.

The cover note, word for word

To: Credit desk
From: Operations
Subject: Month-end holdings extract, as of 2026-10-30

The month-end extract is attached as project_holdings.csv.

Control totals for the book:

Number of bonds:	206
Total face:	507,250,000 dollars
Total market value:	509,928,477.50 dollars

Market value is face times clean price plus accrued interest, divided by 100.
All positions are priced as of 2026-10-30.
Please tell us if the extract does not agree with these totals.

The three control totals are given to you. Test the file against them.

The note is also in the course repo, as data/project_cover_note.txt.

What you hand in

One thing: the project notebook, completed

- 1 Every code cell filled in and run, top to bottom
- 2 Every decision written down
- 3 The written conclusion of section 9

`week4/day4/course1_project.ipynb`

The worked version

`course1_project_solutions.ipynb`

It is in the same folder, with every answer and a model conclusion.
Open it after you have written your own conclusion.

Before you hand in

Run it all

Restart the kernel and run the whole notebook from the top.

Every check

Every cell runs, and every check prints correct.

The raw file

raw is unchanged: it is the file as it arrived.

Every fix

A markdown cell above it gives the decision and the evidence.

Every chart

A title, and labelled axes with units.

The conclusion

Written, in 250 to 350 words, and a description throughout.

The files, and what each holds

All in the data folder of the course repo. The setup cell loads the first three.

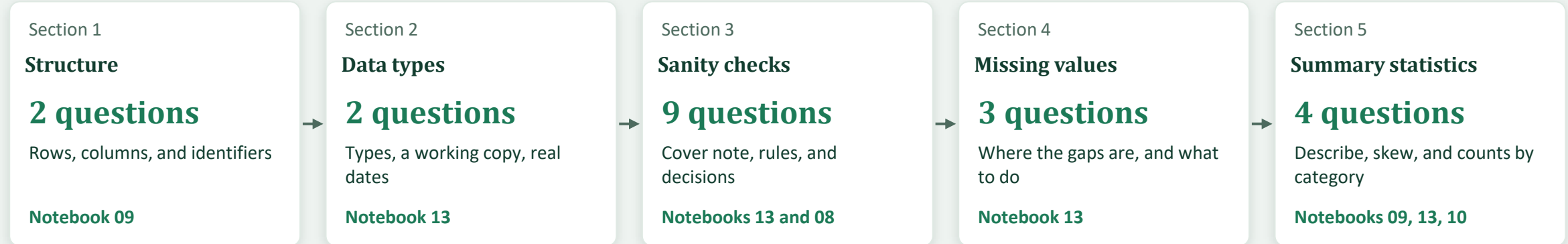
File	What it holds	One row is	As of
project_holdings.csv	The month-end extract from operations	one bond the desk holds, or should be	2026-10-30
benchmark.csv	The benchmark of the book, 821 bonds	one bond, with its weight_pct	2026-09-30
ratings.csv	The rating scale, 18 rows	one rating: score, bucket, and grade	no date
issuers.csv	The issuer table of earlier notebooks	one issuer. The project does not load it	no date
project_cover_note.txt	The note, with three control totals	not a table: a note to read	2026-10-30

The benchmark file is September's. The desk has not been sent an October one, so the two files are a month apart.

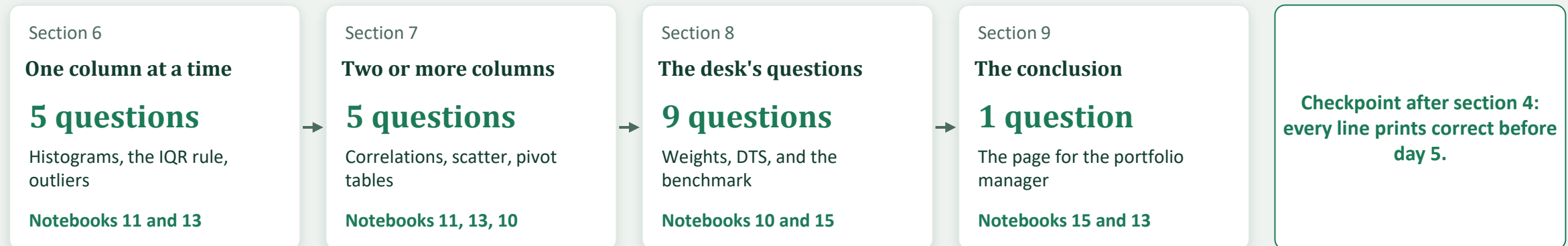
Columns	Unit	Columns	Unit
coupon, yield_pct	percent	par_held, market_value	dollars
oas	basis points	amount_outstanding	dollars, the whole issue
duration, spread_duration	years	dts	spread duration times OAS
price, accrued	per 100 of face	weight_pct	percent, in the benchmark only

Nine sections, 40 questions, two days

Week 4, day 4: sections 1 to 5. The day ends with a table you trust.



Week 4, day 5: sections 6 to 9. The day describes that table.



In green: the notebook that taught the method. The Hints table at the end of the project notebook lists the sections of each.

The rules

1

Work in a copy

raw is never changed. Every fix is made to the working copy, book.

2

State every decision

What you are changing, in how many rows, and on what evidence.

3

Print the headline numbers

Before and after every fix, with headline() from the setup cell.

4

Describe, never recommend

Nothing recommends, ranks, or forecasts a bond, issuer, or sector.

5

Charts are finished charts

A title, a label on each axis, and the unit.

6

The notebook runs from top to bottom

Restart and Run All before you hand it in.

Rule 3, as code

```
headline('Before', book)

# the fix goes here, made to book

headline('After', book)
```

What headline() prints

```
Before
rows: ...
face: ... dollars
market value: ... dollars
mean OAS: ... basis points
```

All names are invented

No name refers to a real company. Overweight and underweight name the sign of a subtraction, and nothing more.

How the checks work

A question, its empty cell, and its check cell

```
# your code here
answer_1_1 = ...

# the next cell is written for you
check('Question 1.1', answer_1_1, expected)
```

check() prints	When
Question 1.1: not attempted yet	the answer is still ...
Question 1.1: not yet. You have	your answer is different
Question 1.1: correct	your answer matches

A check tests the number

It does not test your reasoning. Charts and written answers have no check: each of those questions says what a good answer includes.

When a check disagrees

- 1

Read the question again
The unit, the rounding, and which table: raw or book.
- 2

Find the expected decision
The question states the decision its check expects.
- 3

Print the headline numbers
Compare them with the control totals of the cover note.
- 4

Made a different decision?
State it in a markdown cell. The check will disagree.
- After section 4, a checkpoint tests book against the cover note.

Using an AI assistant

An AI assistant is allowed, used the way the course taught: four steps.



An answer you cannot check is not finished.

Also allowed

Every earlier notebook of the course, and your own notes
The documentation of pandas, matplotlib, and seaborn

Nothing from an employer

Nothing from an employer goes into an AI tool or into Colab. The course data is synthetic for that reason.

The written conclusion

Question 9.1: the page for the portfolio manager, in 250 to 350 words

Part 1

Five numbers

The five figures a reader needs first, each with its unit.

Part 2

Concentrations

Where the market value and the DTS sit: by sector, grade, and issuer.

Part 3

Data quality

Each problem, in how many rows, and what was done. What was found clean.

Part 4

The benchmark

Differences in weight by sector and rating bucket, and the two file dates.

Part 5

Open items

What the operations team needs to confirm or send.

Describe. Do not recommend, rank, or forecast.

Overweight and underweight mean a positive or a negative difference in weight, and nothing else.

There is no check cell for this question.

A good conclusion

Can be read without the notebook

Has a number in almost every sentence

Gives the unit of each number

Holds nothing that a cell above does not support

What you can do now

- Write Python: variables, loops, functions, classes, and reading an error
- Hold a portfolio in pandas: filter it, group it, join it, and summarize it
- Draw finished charts: a title, labelled axes, and units
- Test a new file before you trust it, and state every decision
- Write up what the data shows, without a recommendation in it

Course 2 Bond Math: pricing, yield, duration, and convexity, built by hand and checked against Excel